

Obstacles for a Llama

Inyamanswa llama irashaka kunyura ahantu bita Andean plateau. Ifite map ya plateau iri muri form ya grid ifite $N \times M$ square cells. Rows za map zihera kuri 0 kugera kuri $N - 1$ uturutse hejuru kugera hasi, kandi columns nazo zifite ikiminyetso guhera kuri 0 kugera kuri $M - 1$ uherye ibumoso ujya iburyo. Cell ya map muri row i na column j ($0 \leq i < N, 0 \leq j < M$) yerekanwa nka (i, j) .

Llama yize ikirere cy' iyo region ibona ko cells ziri muri row runaka ya map zifite **temperature** imwe kandi cells ziri muri column runaka ya map zifite **humidity** imwe. Llam yaguhaye arrays ebyiri zirimo imibare: T na H zifite length N na M respectively. $T[i]$ ($0 \leq i < N$) yerekana ubushyuhe (temperature) bwa cell iri muri row i , kandi $H[j]$ ($0 \leq j < M$) yerekana humidity ya cells ziri muri column j .

Llama yize ku bimere biri muri plateau ibona ko cell (i, j) **idafite ibimera** mu gihe temperature yayo iruta humidity yayo, bishatse kuvuga igihe $T[i] > H[j]$.

Llama ishobora kunyura muri plateau ikurikije **Inzira zemewe**. Inzira yemewe ni urutonde rwa cells zitandukanye zikurikije ibi bikurikira:

- Buri tsinda rya cells zikurikiranye zihuriye ku ruhande rumwe.
- Cells zose ziri muri path nta bimera zifite.

Akazi kawe ni ugusubiza ibibazo Q . Kuri buri kibazo, urahabwa imibare ine: L, R, S , and D . Ugomba kwerekana niba hari path yemewe aho:

- Path itagirira kuri $(0, S)$ kandi ikaragirira kuri cell $(0, D)$. Birazwi neza ko $(0, S)$ na $(0, D)$ zidafite ibimera.
- Ibimera byose biri muri path biri muri columns L na R , inclusive.

Implementation Details

Function yambere ugomba gukora n' iyi:

```
void initialize(std::vector<int> T, std::vector<int> H)
```

- T : array ifite length N yerekana temperature yo muri buri row.
- H : array ifite length M yerekana humidity yo muri buri column.
- Ihamagarwa inshuro imwe kuri buri test case, mbere yuko ihamagara `can_reach`.

Function ya kabiri ugomba gukora ni iyi:

```
bool can_reach(int L, int R, int S, int D)
```

- L, R, S, D : imibare yerekana ikibazo.
- Iyi function ihamagarwa inshuro Q .

Iyi function igomba ku returninga `true` igihe hari inzira ishobora kuva kuri cell $(0, S)$ ikagera kuri cell $(0, D)$, aho cells ziri muri iyo nzira zibarizwa muri columns L to R , inclusive.

Constraints

- $1 \leq N, M, Q \leq 200\,000$
- $0 \leq T[i] \leq 10^9$ kuri buri i aho $0 \leq i < N$.
- $0 \leq H[j] \leq 10^9$ kuri buri j aho $0 \leq j < M$.
- $0 \leq L \leq R < M$
- $L \leq S, D \leq R$
- Cells $(0, S)$ na $(0, D)$ zidafite vegetation.

Subtasks

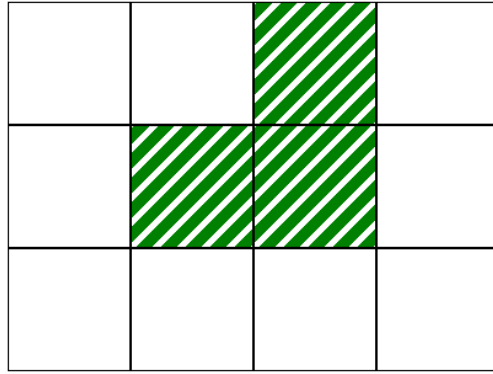
Subtask	Score	Additional Constraints
1	10	$L = 0, R = M - 1$ kuri buri kibazo. $N = 1$.
2	14	$L = 0, R = M - 1$ kuri buri kibazo. $T[i - 1] \leq T[i]$ kuri buri i aho $1 \leq i < N$.
3	13	$L = 0, R = M - 1$ kuri buri kibazo. $N = 3$ and $T = [2, 1, 3]$.
4	21	$L = 0, R = M - 1$ kuri buri kibazo. $Q \leq 10$.
5	25	$L = 0, R = M - 1$ kuri buri kibazo.
6	17	Ntazindi constraints.

Example

Nk' urugero:

```
initialize([2, 1, 3], [0, 1, 2, 0])
```

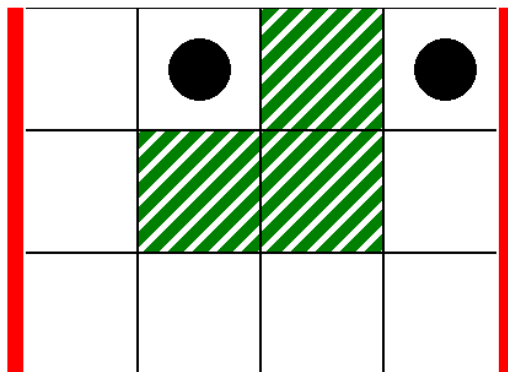
Ibi bijyanye na map iri muri iyi foto, aho cells zisa umweru zidafite ibimera:



Nk' ikibazo cyambere, reba iyi call:

```
can_reach(0, 3, 1, 3)
```

Ibi bijyanye n' ifoto yerekanwe hasi, aho umurongo munini umanuka werekana columns ziva kuri $L = 0$ zikagera kuri $R = 3$, kandi disk(ikizero) y' umukara yerekana cells zatagiriweho n' izaragiriweho:



Iki gihe, llama ashobora kuva kuri (0,1) akagera kuri (0,3) anyuze muri iyi nzira(path):

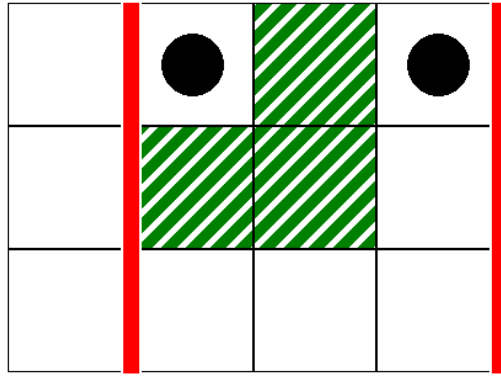
(0,1), (0,0), (1,0), (2,0), (2,1), (2,2), (2,3), (1,3), (0,3)

Bityo rero, iyi call igomba kureturninga `true`.

Nk' ikibazo cya kabiri, tekereza kuri iyi call:

```
can_reach(1, 3, 1, 3)
```

Ibi bijyanye n' iyi foto:



Muri iki gihe, ntanzira ihari iva kuri cell $(0, 1)$ ikagera kuri cell $(0, 3)$, aho cells ziri muri path ziba ziri muri columns 1 kugera kuri 3, inclusive. Bityo rero, iyi call igomba kureturninga `false`.

Sample Grader

Input format:

```
N M
T[0] T[1] ... T[N-1]
H[0] H[1] ... H[M-1]
Q
L[0] R[0] S[0] D[0]
L[1] R[1] S[1] D[1]
...
L[Q-1] R[Q-1] S[Q-1] D[Q-1]
```

Hano, $L[k]$, $R[k]$, $S[k]$ and $D[k]$ ($0 \leq k < Q$) byerekana values za buri call to `can_reach`.

Output Format:

```
A[0]
A[1]
...
A[Q-1]
```

Hano, $A[k]$ ($0 \leq k < Q$) ni 1 iyo call `can_reach(L[k], R[k], S[k], D[k])` yariturninze `true`, na 0 iyo byagenze ukundi.